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The Economics of Organizational Architecture

by James Brickley, Clifford Smith, and Jerold Zimmerman, University of Rochester

Francis Baring with his brother John established Barings Bank in London in 1762. Their bank prospered by facilitating international trade. Barings helped finance the British effort in the American Revolutionary War, and thereafter Barings credit reopened trade with the United States. In 1803, Barings helped the U.S. finance the Louisiana Purchase; it also helped finance Britain's wars against Napoleon. In 1818, the bank's influence was such that Duc de Richelieu observed, "There are six great powers in Europe: England, France, Prussia, Austria, Russia, and Barings Brothers."

The bank was shaken in 1890 when loans that it had made in Argentina defaulted. But it survived with the help of a bailout engineered by the Bank of England. The family rebuilt the bank and, although it never regained its former preeminence, Barings remained a gilt-edged institution run largely by members of the family, and owned primarily by a charitable foundation. In the first half of the 1990s, its influence grew significantly, in part due to its substantial Far East securities business.

In late February of 1995, Barings' board of directors met to review the 1994 results. The bank had a small rise in profits—a quite reasonable result in what had been a dreadful year for most of its competitors. One big contributor to those results had been a very profitable securities operation in Singapore. But that afternoon, things changed dramatically. The Singapore office trading star, Nick Leeson, unexpectedly walked out of the office and disappeared. As senior management examined the bank's records, it became clear that something was very wrong.

In principle, Leeson was engaged in a simple arbitrage operation. He traded futures contracts on the Nikkei 225 (the main Japanese stock market index) on both the Osaka Stock Exchange and the Singapore International Monetary Exchange (SIMEX). He should have been able to lock in a virtually riskless profit by selling the security on the exchange

with the higher price while simultaneously buying it on the exchange with the lower price. In this arbitrage business, although Barings would accumulate large positions on both exchanges, those securities it bought and those it sold should balance; there was supposed to be no net exposure.

Yet what management found as they reviewed the bank's records was that Leeson had bought securities in both markets; in effect, he had made an enormous bet that the Nikkei would rise. But it had fallen, and now the solvency of the bank was threatened.

How could this have happened? It appears that Leeson circumvented the bank's internal controls. The Singapore branch was small and Leeson had effective authority over both trading *and* the firm's back-office systems. He used that power to conceal losses and disguise the true nature of his activities—and thus he was able to "cook the books." For example, he apparently told senior management that a number of his trades were on behalf of clients and not the bank. And the bank's internal control systems did not expose the deceit.

By early March, the bank's aggregate losses totaled \$1.4 billion. Leeson, who had been arrested by German police at the Frankfurt Airport, was fighting extradition to Singapore. And Barings, Britain's oldest merchant bank, had been sold to ING, the large Dutch financial institution, for £1. Thus, Barings' owners had lost their entire investment.

The collapse of Barings was ultimately caused by a poorly designed organization. As the *Wall Street Journal* noted:

*This article draws on Chapters 1, 8, and 15 of *Managerial Economics and Organizational Architecture* (McGraw-Hill/Irwin, 2004). We owe a significant debt to our colleagues, Michael C. Jensen and William H. Meckling. Their work has profoundly affected our thoughts on these issues.

*What is emerging from the documents and from interviews with current and former Barings executives is a fatally flawed organization: one that ignored at least several warning signs going back not just weeks and months, but years; one that so wanted to ensure the continuation of profits from Singapore—which boosted bonuses—that it was reluctant to impose tight controls; one that had a deeply split staff, which ultimately may have contributed to its downfall.*¹

Three general aspects of the bank's organization contributed to the failure: (1) the broad range of authority and responsibilities granted to Leeson; (2) gaps in the bank's systems for evaluating, monitoring, and controlling its employees; and (3) aspects of the firm's compensation system. Let's examine each in more detail.

First, Leeson had responsibility for both proprietary and customer trading as well as effective control of the settlement of trades in his unit. This broad assignment of decision-making authority created the opportunity to circumvent the bank's internal controls. As the *Financial Times* noted:

*In Singapore, Mr. Leeson was in the process of settling transactions as well as initiating them. A watertight line between dealing and operational responsibility, crucial to internal control, was missing.*²

In reaction to the Barings collapse, SIMEX changed its rules to require that member firms ensure that proprietary traders not handle customer business and that the head of the dealing section not take charge of the settlement process.

Second, Leeson compromised the firm's performance evaluation system. He misrepresented trades for the bank as customer trades and hid losses. A better designed and executed monitoring system would have identified these problems before the solvency of the institution was threatened.

Third, the bank's compensation system encouraged Leeson to speculate while providing senior managers with limited incentives to exercise tight control over their star trader. Barings traditionally had paid out approximately 50% of gross earnings as annual bonuses. Yet a system where managers participate in annual profits—but not in losses—can encourage excessive risk-taking. This incentive can be most pronounced when a small bet loses and the employee tries to make it up by doubling the bet. If this second bet also loses, there can be a strong incentive to double up again and “go for broke.”

This tale of Barings' untimely end serves to illustrate a critical idea—namely, that the organization of the firm matters.³ A poorly designed organization can result in lost profits and even in the failure of the institution. And competition in the marketplace provides strong pressures for more efficiently designed organizations. A company that clings to an inefficient, high-cost organizational design is eventually forced either to adapt or close.

With the benefit of hindsight, it seems easy to identify aspects of Barings' organization that, if changed, might have prevented this debacle. But the critical managerial question is whether one could reasonably have identified the potential problems *before* the fact and designed a more productive organization. We believe the answer to this managerial question is a resounding yes.

“

Although competition tends to produce efficiently organized firms over the longer run, uncritical experimentation with the organizational innovation *du jour* can expose the firm to an uncomfortably high risk of failure.

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We are not, of course, the first to recognize the importance of corporate organization or to offer advice on how to improve it. The business section of any good bookstore displays an array of titles featuring many of the following: *Benchmarking, Empowerment, Total Quality Management, Reengineering, Outsourcing, Teaming, Corporate Culture, Venturing, Matrix Organizations, Just-In-Time Production, and Downsizing*. The authors of all these books would strongly agree with the proposition that the firm's organization and the associated policies chosen by management can have profound effects on performance and firm value. And they all buttress their recommendations with selected stories of firms that followed their advice and achieved fabulous successes.

The problem with such approaches, however, is that each tends to focus on a particular facet of the organization,

1 M. Branchli, N. Bray, M. Sesit, “Barings PLC Officials May Have Been Aware of Trading Position,” *Wall Street Journal*, March 6, 1995, p. 1.

2 “The Box That Can Never Be Shut,” *Financial Times*, February 28, 1995, p. 17.

3 It is an idea that most economists tend to lose sight of. Economists have given surprisingly little attention to questions of the firm's organization. In traditional economic analysis, the firm is generally characterized simply as a “black box” that transforms inputs (like labor, capital, and raw materials) into outputs. Although more attention has been given in recent years to the ways in which companies are organized, there has been little effort devoted to synthesizing the material in an accessible way that emphasizes the managerial implications of the analysis.

ECONOMIC DARWINISM: THE CASE OF FRANCHISING

As described in Charles Darwin's *The Origin of Species*, natural history illustrates the principle of "survival of the fittest." In industry, we see Economic Darwinism in operation as competition weeds out those organizations that fail to adapt. There is an important difference, however, between these two evolutionary processes. In the biological systems that Darwin studied, the major forces at work were random mutations in organisms and shocks to the external environment. Within economic systems, voluntary purposeful changes are also possible. Indeed, managers are quick to imitate the organizational innovations of other successful firms.

As one illustration of Economic Darwinism, consider the rise of franchising in the 1950s. In those days, most quick-service restaurants in the U.S. were local "mom and pop" businesses. Similarly, small independent businesses tended to dominate industries such as motels, convenience stores, and auto repair. Individual businesses in these industries competed with one another and the most profitable tended to survive. Nevertheless, there was room for improvement. In particular, the quality across businesses was not uniform and consumers bore risks in trying new establishments. Vacationers, for example, were not sure what to expect when patronizing out-of-town restaurants and motels, and the importance of these issues increased as improved highways expanded travel.

Entrepreneurs responded to this opportunity by developing a relatively untried form of organization—franchising. For example, during the 1950s Ray Kroc, Bill Rosenberg, and Kemmons Wilson founded McDonald's, Dunkin' Donuts, and Holiday Inn, respectively. (Other franchise companies formed during this period include H&R Block, Midas Muffler, and Pizza Hut, to name a few.) The idea in each case was to establish a national chain to provide customers with uniform quality and service at all outlets.

One important advantage of franchising over conventional corporate ownership is that franchisees are more highly motivated than typical corporate managers to increase long-term sales and profits because franchisees pay for the franchise, keep much of the profits, and can sell the franchise. At the same time, franchising retains the "corporate" advantages of group affiliation for national advertising, quantity discounts for input purchases, and quality control.

The success of franchising has been immense. There are now over 500,000 franchised units operating in this country with sales of nearly \$800 billion, representing about one-third of the nation's retail sales. Franchising now dominates industries such as fast food, auto repair, and motels. Thus, many small independently owned companies were either driven out of business by franchise companies or joined franchise systems themselves.

whether it be quality control or worker empowerment or the compensation system, to the virtual exclusion of all others. As a consequence, the suggestions offered by the business press are often mutually inconsistent. These publications tend to offer little guidance as to which tools are most appropriate in which circumstances; the implicit assumption of most is that their technique can be successfully adopted by all companies. This assumption, however, is usually wrong.

For example, if the cover article in the next *Businessweek* reports an innovative inventory control system at Toyota, managers around the globe will see it and ask, "Would that work in my company, too?" Undoubtedly the managers with the strongest interest in trying it will be those in firms with current inventory problems. Some will achieve success, but others may experience disastrous results caused by unintended, though largely predictable, organizational

"side-effects" (like Leeson's unchecked appetite for risk-taking).

Although competition tends to produce efficiently organized firms *over the longer run*, uncritical experimentation with the organizational innovation *du jour* can expose the firm to an uncomfortably high risk of failure. For this reason, it is important to be able to analyze the likely consequences of a contemplated change in organization for the entire firm.

In contrast to the approach of most business bestsellers, then, we seek to provide a systematic framework for addressing such issues—an approach that can be applied consistently in analyzing organizational problems and designing more effective organizations. This framework identifies three critical aspects of corporate organization:

- (1) the assignment of decision rights within the firm;

(2) the structure of systems to evaluate the performance of both individuals and business units; and

(3) the methods of rewarding individuals.⁴

We introduce the term *organizational architecture* to refer specifically to these three key aspects of the firm. Although we could instead have simply used “organization” to refer to these three corporate features, common usage of that term refers only to the hierarchical structure (that is, reporting relationships and decision right assignments), but it ignores the performance evaluation and reward systems. We thus use “organizational architecture” to help focus attention on all three of these key aspects of the organization.

Stated as briefly as possible, our argument is that successful firms assign decision rights in ways that effectively link decision-making authority with the relevant information for making good decisions. When assigning these decision rights, however, top management must also ensure that the company’s performance evaluation and reward systems provide decision-makers with appropriate incentives to make value-increasing decisions.

Our approach to corporate organization is “integrative” in the sense that it draws on a number of disciplines: accounting, finance, marketing, management, and political science. But what also distinguishes this approach most clearly from that of the bestsellers is its central reliance on the basic principles of economics. In essence, economics provides a theory of how individuals make choices. For example, in designing organizations, one must always keep in mind that individuals respond to incentives. Managers and employees can be incredibly resourceful in devising methods to exploit the opportunities they face. This also means, however, that when their incentives are structured improperly, they can act in ways that reduce firm value. For this reason, in choosing corporate policies, it is critical that managers anticipate potential responses by customers, suppliers, or employees that might produce undesirable outcomes. Failure to do so invites individuals to “game” the system and can result in the complete failure of well-intentioned policies.

More generally, this approach to corporate organization uses the basic tools of economics to examine the likely effect on firm value of decisions such as centralization versus decen-

tralization, the bundling of tasks into specific jobs within the firm, the use of objective versus subjective performance measures, compensating employees through fixed versus variable (or “incentive”) compensation, and retaining activities within the firm versus outsourcing. In sum, we examine how managers can design their organizational architecture to motivate individuals to make choices that increase firm value.

The Fundamental Problem

The primary goal of any economic system is to produce the output customers want at the lowest cost possible. The challenge of discovering customer demands while reducing costs, both for economic systems and *within* individual firms, is complicated by the fact that important information for economic decision-making is generally held by many different individuals. Furthermore, this information is often expensive to transfer (that is, the information is *specific* as opposed to *general*).⁵ For example, a scientist is likely to know more about the potential of a new research project than people higher up in the firm. Similarly, a machine operator may know more about how to use a particular machine than his supervisor. In both cases, a decision-making process that requires the communication of such information to headquarters for approval is likely to be cumbersome, resulting in many lost opportunities.

A second complication is that decision makers might not have appropriate incentives to make value-increasing decisions even if they have the relevant information (that is, there are *agency problems*).⁶ For example, a scientist might want to complete a research project out of scholarly interest even if the project is unprofitable. Similarly, machine operators might not want to use machines efficiently if this means more work for them.

In sum, the principle challenge in designing both firms and economic systems is to maximize the likelihood that decision-makers have (1) the relevant information to make good decisions and (2) the incentives to use the information productively.

The Right Architecture Is Built into Markets. The price system helps solve agency problems in markets. In market economies, individuals have private property rights. If a person

4 We thank Mike Jensen and Bill Meckling for impressing upon us the importance of these three features of organizations. See M. C. Jensen, “Organization Theory and Methodology,” *The Accounting Review* 58 (1983) 319-339; and M. C. Jensen and W. H. Meckling, “Specific and General Knowledge and Organizational Structure,” in *Contract Economics*, Lars Werin and Hans Wijkander, eds., Basil Blackwell Ltd., Oxford, U.K., 1992, chapter 9. These three features are identified throughout the economics and management literature on organizational design. For example, see P. Milgrom and J. Roberts, *Economics, Organization & Management*, Prentice-Hall, Englewood Cliffs, NJ, 1992. A representative example, from the management literature is D. Robey, *Designing Organizations*, Richard D. Irwin Inc., Burr Ridge, IL, 1991.

5 Our discussion in this section closely parallels M. C. Jensen and W. H. Meckling, “Specific and General Knowledge and Organizational Structure,” in *Contract Economics*, Lars Werin and Hans Wijkander, eds., Basil Blackwell Ltd., Oxford, U.K., 1992, chapter 9.

6 In this corporate context, the agency problem refers to conflicts of interest between shareholders and corporate managers and employees that reduce the value of the firm. For the original formulation of this problem, see Michael C. Jensen and William H. Meckling, “Theory of the Firm: Managerial Behavior, Agency Costs and Ownership Structure,” *Journal of Financial Economics*, Vol. 4, No. 4 (October 1976).

owns a piece of property, an automobile, or a building, that person decides how to use it. Furthermore, this right can be sold to others. If someone else knows how to make better use of the resource, the owner can sell it and keep the proceeds. Owners thus have strong incentives to use resources productively because they bear the wealth effects of their decisions.

Hence, the market provides an architecture that promotes efficient resource use. First, through market transactions, decision rights for resources are rearranged so that they are held by individuals with the relevant specific knowledge. Second, the market provides a mechanism for evaluating and rewarding the performance of resource owners—owners bear the wealth effects of their decisions. This mechanism generates strong incentives to take efficient actions.



Organizational architecture is like a “three-legged stool”: changing one of the three legs without careful consideration of the other two is typically a mistake.



Organizational Architecture Must Be Designed for Firms. One especially valuable feature of markets is that the organizational architecture is created *automatically* with little conscious thought or human direction. Within firms, however, there are no automatic systems either for assigning decision rights to individuals with information or for motivating individuals to use information to promote the firm’s objectives. Organizational architecture must be created.

Although transfer prices are used to allocate some resources within firms,⁷ most resources are allocated by administrative decisions. For example, the CEO typically transfers a manager from one division to another by a simple command. Similarly, how a plant is used can be changed by administrative order. The top management of an organization must decide how to assign the decision rights among employees. For instance, does the top management make most major decisions or are these decisions delegated to lower-level managers? As another example, are machine operators permitted to deviate from procedures outlined in company manuals?

Through this assignment of decision rights, employees are granted authority over how to use company resources.

But employees are not owners—they cannot sell company property and keep the proceeds. Thus, employees have fewer incentives to worry about the efficient use of company resources than the owners. To help control such agency problems, the firm must develop a *control system*. That is, it must develop both (1) performance evaluation systems and (2) reward systems that align the interests of decision-makers with those of the owners.

It is important to recognize, moreover, that *the control system must be consistent with the assignment of decision rights*. While the optimal control system depends on how decision rights are assigned in the firm, the assignment of decision rights depends in part on management’s ability to devise an effective control system. For example, if employees are given greater decision-making authority, it will usually make sense to develop new performance evaluation and reward systems that tie the employees’ compensation more closely to the new performance measures. By the same token, if a firm adopts a compensation plan to motivate employees, it is important to ensure that workers have decision rights that enable them to respond to such incentives.

In this sense, the three components of organizational architecture—the assignment of decision rights, the performance evaluation system, and the reward system—are all highly interdependent. Indeed, organizational architecture is like a “three-legged stool”: changing one of the three legs without careful consideration of the other two is typically a mistake—they must be designed together to keep the stool level.

Once the firm grows beyond a certain size, the CEO is unlikely to have the relevant information for all major decisions. Consequently, this CEO has three basic choices in designing the organization.

First, the CEO can continue to make most major decisions even if he does not have the relevant information. The benefits of this choice are limited agency problems and hence little need for a detailed control system.⁸ The costs of such centralization, however, will come in the form of suboptimal decisions resulting from the lack of specific information.

Second, the CEO can attempt to acquire the relevant information to make better decisions. But obtaining and processing the relevant information can be very time consuming and costly.

Third, the CEO can decentralize decision rights to individuals with better information. This choice assigns

7 One reason for the existence of firms is that coordinating related transactions through the market price mechanism can be expensive. Nevertheless, some firms use internal transfer prices in resource allocation decisions. See our discussion of “Transfer Pricing” later in this issue.

8 We assume for simplicity he is also the sole owner of the business. In small firms, top management and owners are often the same. In large firms, however, the owners (the shareholders) delegate most decision rights to the board of directors and the CEO. These parties are charged with developing the organizational architecture for the firm. For purposes of this discussion, we treat top managers and owners as the same.

WHEN THE LEGS OF THE STOOL DON'T BALANCE

A plane owned by a major airline company was grounded for repairs at one airport, with the nearest qualified mechanic stationed at another. The decision right to allow the mechanic to work on the airplane was held by the manager of the second airport. But because that manager's compensation was tied to meeting his own budget rather than to the profits of the overall organization, he refused to send the mechanic immediately to fix the plane because the mechanic would have had to stay overnight at a hotel and

the hotel bill would have been charged to the manager's budget. The mechanic was dispatched the next morning so that he could return the same day. Thus, a multimillion-dollar aircraft was grounded, costing the company thousands of dollars, so that a manager's budget would avoid a \$100 hotel bill.

Suggested by M. Hammer and J. Champy, *Reengineering the Corporation*, Harper Business, New York, 1993.

THE ORGANIZATIONAL ARCHITECTURE OF CENTURY 21

Century 21 International is the largest real estate firm in the world, operating throughout the United States and in ten countries, including Japan, the United Kingdom, and France. Given the geographic and cultural diversity of its operations, it would not make sense for the U.S. head-quarters to make all major decisions. Such centralized decision-making would be especially problematic for the international operations, where laws and cultures can be far different than those in the U.S. To quote Century 21's management: "We provide the international regions with whatever knowledge we possess on how they can help their franchisees develop better offices. What they use is basically up to them and will reflect their housing market and real estate traditions. We allow our master subfranchisors a great deal

of flexibility in running their regions, and internationally we want them to be able to accommodate their services to their culture."

Decentralized decision-making, however, requires a control system that promotes productive effort. At Century 21, most of the local operators are franchisees. Franchisees are essentially owners of their units and keep a large share of their units' profits. This ownership provides strong incentives to increase sales and Century 21's value. But Century 21 also reserves the right to terminate individual franchises that fail to maintain acceptable levels of service.

Suggested by C. Shook and R. L. Shook, *Franchising: The Business Strategy that Changed the World*, Prentice Hall, NJ, 1993.

decision-making authority to employees with the relevant specific information. But delegating decision rights gives rise to agency problems, which means that control systems must be developed.

Another potential drawback of decentralization is the cost of transferring information *from top management to the decentralized decision-makers* in coordinating efforts throughout the firm. (In the story of the grounded airplane above, the CEO would certainly have sent the mechanic had the matter been brought to his attention.)

Of course, managers can choose a mix of these basic alternatives. For example, top managers are likely to choose to retain some decisions while delegating others. The optimal choice, as we discuss below, depends primarily on the business

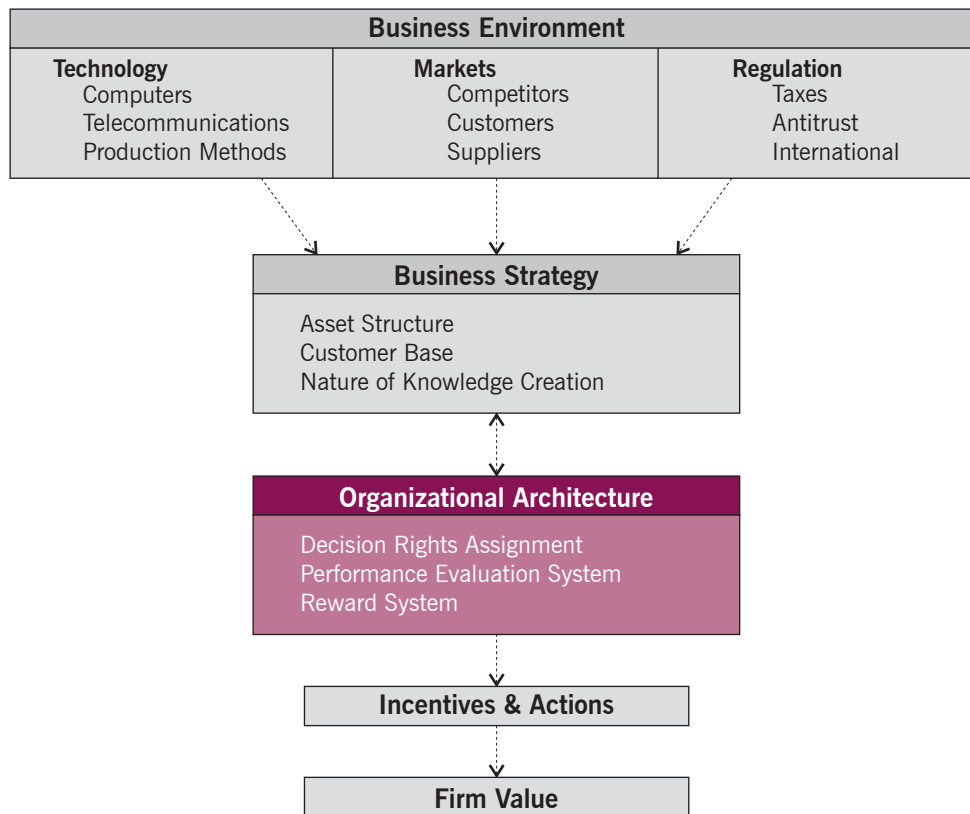
environment and strategy of the firm. In some cases—especially smaller firms in relatively stable industries—top managers are likely to have most of the relevant information for decision-making, and thus decision rights are more likely to be centralized at headquarters. In other cases—especially larger firms experiencing rapid change—top managers and their corporate staff often will not be in the best position to make many decisions. And, in such cases, decision rights are more likely to be decentralized with corresponding control systems put in place.

Determinants of Organizational Architecture

As suggested above, then, the optimal architecture will be different for different companies. Such structural differences

Figure 1

Determinants of Organizational Architecture



are not random, however, but vary in *systematic* ways with differences in certain underlying characteristics of the companies themselves. To illustrate the point, companies operating in the same industry tend to develop similar organizational architectures. Furthermore, if an important aspect of an industry changes, many companies in that industry end up making major changes in decision rights and internal control systems.

In Figure 1, we summarize those factors that are likely to be most important in designing the optimal architecture for a given firm. At the top of the figure are three key aspects of the firm's *external* business environment: *technology*, *markets*, and *regulation*.⁹ For any firm, these three factors—(1) the technol-

ogy that affects both its products, its methods of production, and its information systems, (2) the structure of its markets (competitors, customers, suppliers, etc.), and (3) the regulatory constraints on its activities—are likely to have the greatest influence on the business strategy. By business strategy, we mean the answers to a broad set of questions: What are the firm's primary goals, non-financial as well as financial? What products and services is it providing, and to what customers? And what are the firm's sources of comparative advantage in so doing?

Take the case of AT&T before it was broken up in the early 1980s. Regulation dictated many aspects of the firm's business strategy—what services it could offer, what customers it could serve and how much it could charge them, and what technologies it could employ. After the breakup of AT&T and the deregulation of the telecommunications industry, both the Baby Bells and the new AT&T were forced to devise new

9 Although we can point to cases that contradict this generalization, Figure 1 treats the business environment as *exogenous*, or beyond the managers' control. By contrast, the business strategy and organizational structure are *endogenous* variables—that is, they are policy choices made by the management *in response* to the business environment.

By treating the business environment as exogenous, our analysis does not focus on potential feedback effects among the environment, business strategy, and organizational structure. Consider, for example, how the organizational structure at Microsoft encourages innovation that, in turn, alters the basic technology facing the firm. Large firms also often have political power that can be used to influence government regulation. Although these types of feedback effects can sometimes be important, in most circumstances

managers essentially must take the business environment as outside their control.

INFORMATION TECHNOLOGY AND CHANGES IN ORGANIZATIONAL ARCHITECTURE AT CYPRESS SEMICONDUCTOR

By allowing top management to communicate more directly with lower-level employees, computer technology makes it less expensive for managers to control and coordinate their actions. It thus has reduced the demand for middle managers, who traditionally have played an important role in transmitting information from the top of the corporation to lower levels.

An example of this took place at Cypress Semiconductors. T. J. Rodgers, CEO of Cypress, uses a computer system to track the daily objectives of every employee in the company. The company has essentially no middle management. As noted in a *Fortune* article,

The computer system allows the CEO to stay abreast of

every employee and team in his fast-moving organization. Each employee maintains a list of ten to 15 goals like “Meet with marketing for new product launch,” or “Make sure to check with Customer X.” Noted next to each goal is when it was agreed upon, when it’s due to be finished, and whether it’s finished or not.

This way, it doesn’t take layers of expensive bureaucracy to check who’s doing what, whether someone has got a light enough workload to be put on a new team, and who’s having trouble. Rodgers says he can review the goals of all 1,500 employees in about four hours, which he does each week.

Suggested by Brian Dumaine, “The Bureaucracy Busters,” *Fortune*, June 17, 1991, p. 46.

strategies to provide new products, serve new customer bases, and develop new pricing structures.

To return to Figure 1, the ultimate goals of the firm, as reflected in its business strategy, in turn affect its optimal organizational architecture. As the celebrated architect Louis H. Sullivan once observed, “Form ever follows function.” Applying the same principle to industry, we see that significant changes in corporate business strategies typically call for major changes in decision-making authority, performance measures for evaluating managers and employees, and management incentive compensation systems.

Returning to our telecommunications example, in the early 1980s a regulated AT&T faced little competition or pressure for technological innovation. It operated in a reasonably stable environment—one where it made sense for a huge formal bureaucracy to make the most important decisions “from the top down.” Since the breakup of the company, however, the telecommunications industry has experienced almost continuous upheaval, with deregulation, increased competition, and rapid technological change. And, in 1992, after a nearly decade-long series of incremental moves toward decentralization, AT&T established a large number of fairly autonomous profit centers run by managers on pay-for-performance plans tied to economic value added (or EVA®).¹⁰

In general, deregulation causes firms to decentralize decision-making and to redesign their performance evaluation and reward systems to give lower-level workers stronger incentives to make the best use of their expanded decision rights. The impact of technological change on organization, however, is less clearcut than the effect of deregulation. Take the case of advances in information technology. More powerful corporate management information systems (MIS) could lead to greater centralization of decision-making by increasing the ability of corporate headquarters to monitor line managers’ performance. But such advances could also allow for greater decentralization by allowing corporate staffs to implement more sophisticated and effective internal control systems. Advances in corporate MIS can also facilitate decentralization by channeling important information in the opposite direction: namely, from the corporate center to line managers and employees.

Another pronounced effect on organizational architecture has come from increases in foreign competition. Before the 1980s, many large American companies (for example, ITT, IBM, General Motors, Eastman Kodak, and Xerox) faced limited competition in their product markets. Indeed, many of these companies had substantial market power, and thus they had little incentive to focus on rapid product development, high-quality production, or competitive pricing. Quite predictably, they had become highly bureaucratic with very centralized decision-making and limited incentive compensation.

¹⁰ For an introduction to EVA®, see the next article in this issue, Joel Stern, Bennett Stewart, and Don Chew, “The EVA Financial Management System.”

LESSONS FROM ITT

In 1984, ITT Corporation was the largest manufacturer of telecommunications equipment in the world, operating in over 80 different countries. It was also broadly diversified with operations in industrial and consumer products, insurance, automotive parts, telephone service, natural resources, food processing and utilities. ITT, however, faced a variety of market pressures that made 1984 an especially trying year. To quote *Moody's Handbook of Common Stocks* (Winter 1984-85):

ITT's telecommunications operations continued to suffer from soft market conditions in the U.S., a personal computer glut and competitive pricing. European operations are being hurt by the strength of the U.S. dollar.

In 1984, ITT's earnings per share were only \$2.97 (down from \$4.50 a year earlier), dividends were cut by nearly \$1.00 per share, and the company was rumored to be a takeover target.

Part of ITT's problem was that it had become too large and unfocused. Decision-making in the organization was highly formalized and bureaucratic. This system made it difficult for ITT to respond rapidly to changing customer demands and competitive pressures. This inability to act quickly was especially problematic given the dramatic changes occurring in telecommunications and computers. ITT responded by announcing its plan to sell over \$2 billion in assets in order to focus on its core strengths and major lines of business.

As one example of this "asset redeployment program," ITT sold O. M. Scott & Sons Company, the largest producer of lawn-care products in the U.S., to Clayton & Dubilier in a divisional leveraged buyout in December 1986. The buyout was accompanied by organizational changes at Scott that were designed to enhance performance.*

These changes involved all three aspects of organizational architecture:

1. *The Assignment of Decision Rights within the Organization.* After the LBO, managers at Scott were given substantial authority to make and implement decisions, including major capital expenditures. As part of ITT, by contrast, these managers often had to seek approval from a number of levels at ITT headquarters and approval was frequently denied.

2. *The Performance Evaluation System.* The company's new performance-evaluation system placed a heavy emphasis on financial performance. For example, specific targets were established for corporate, divisional, and individual performance that stressed generation of operating cash flow and efficient use of working capital.

3. *The Reward System.* To motivate value-enhancing decisions, coverage under the bonus plan was expanded to include additional managers. Payouts for exceeding performance targets were increased substantially from the old plan. For example, average bonuses as a percent of salary for the ten top managers increased from 10% and 17% in the two years before the buyout to 66% and 39% in the two years after. In addition, managers and employees had substantial financial interests in the firm through stock ownership. After the buyout, they owned 17% of Scott's equity. Before the buyout, managers and employees had no meaningful stock ownership.

These organizational changes at Scott were accompanied by a dramatic increase in operating performance. For example, in the two-year period following the buyout, sales increased by 25%, and earnings before interest and taxes increased by 56%. Moreover, these increases were not caused by slashing expenditures on either R&D or on marketing and distribution. In fact, expenditures in both categories increased, as did spending on capital projects.

*Our discussion of O. M. Scott relies heavily on a paper titled "Organizational Changes and Value in Leveraged Buyouts: The Case of the O. M. Scott & Sons Company," by George P. Baker and Karen H. Wruck, *Journal of Financial Economics*, December 1989. See also the less technical version of the same article that appeared in the *Journal of Applied Corporate Finance*, Vol. 4, No. 1 (Spring 1991).

In the past decade, however, a dramatic increase in foreign competition (notably, from Japanese producers) caused many of these large firms to rethink their basic strategies. In particular, such companies were forced to increase their emphasis on

quality, customer service, and competitive pricing. To accomplish these objectives, such companies also typically found it necessary to redesign their organizations. Many chose to push decision rights lower in the company, where specific

COSTS OF ORGANIZATIONAL CHANGE—NOT A NEW PHENOMENON

“We trained hard, but it seemed that every time we were beginning to form into teams we would be reorganized. I was to learn later in life that we tend to meet any new situation by reorganizing, and what a wonderful method it can be for creating the illusion of progress while producing confusion, inefficiency, and demoralization.”

Petronius Arbiter, 210 BC

knowledge about customer demand was located. These firms also developed new performance evaluation systems focusing more on quality and customer service and increased their use of pay-for-performance compensation plans.

Changing Organizational Architecture

As we have seen, changes in market conditions, technology, or government regulation can change the optimal design of organizations. But organizational change is by no means a costless process.

First there are direct costs. The new structure has to be designed and communicated to employees throughout the organization. Moreover, organizational changes frequently require costly changes in the firm's accounting and information systems. Often what appears to be a straightforward change in the performance evaluation system imposes a major and costly project on the firm's data processing and accounting departments. Literally hundreds of computer programs might have to be changed to alter the accounting and information systems.

Second, and perhaps more important, are indirect costs. Organizational changes are likely to affect some workers positively (say, by increasing their responsibilities and possibilities for rewards) and other workers negatively. Thus, reactions to a change are likely to vary among employees. Dealing with the associated politics of implementing change in an organization can be expensive. Finally, frequent organizational changes can have undesirable incentive effects. For instance, workers have limited incentives to invest in learning new job assignments, devising more efficient production processes, and developing relations with coworkers if they expect to change assignments in the near future. In general, frequent restructuring causes uncertainty about job assignments and will promote actions that focus more on short-run payoffs and less on long-run investments.

Unfortunately, too many firms appear to undertake organizational changes without a careful analysis of the relevant costs and benefits. At any point in time, as we noted earlier, there are a number of prominent management techniques being touted as the key to success. Among the most popular techniques of the 1990s are reengineering, benchmarking, total quality management, broad-banding, worker empowerment, and skill-based pay. Most of these techniques involve fundamental organizational change. For example, many advocates of total quality management recommend delegating decision rights to teams and not paying incentive compensation based on individual performance.

Adopting the most recent “business trend” can get a firm in trouble, however, unless the change is warranted by the actual circumstances facing the firm. As a *Wall Street Journal* article observed, “many companies try management fads, only to see them flop.”¹¹ Furthermore, surveys indicate that a majority of companies are dissatisfied with the results of organizational changes.

Corporate Culture and Organizational Change

One widely cited reason for the failure of such programs for change is that the proposed changes fail to mesh with, and are eventually defeated by, the existing *corporate culture*. “Corporate culture” is another term that appears often in the organizational design literature. Corporate culture usually encompasses the ways work and authority are organized, the ways people are rewarded and controlled, as well as organizational features such as customs, taboos, company slogans, heroes, and social rituals. But although managers are regularly encouraged to develop high-powered, productive cultures, little concrete guidance is typically provided on how they might accomplish this goal.

¹¹ *Wall Street Journal*, July 6, 1993.

CORPORATE CULTURE AT MARY KAY COSMETICS

Total sales at Mary Kay Cosmetics increased from about \$198,000 in 1963 to over \$613 million in 1993. Mary Kay has built a sales force of 300,000 and has helped to create 74 millionaires (women who have earned commissions of \$1 million or more over their careers).

The organizational architecture at Mary Kay focuses directly on sales. All sales consultants purchase products directly from Dallas at the same price. Rewards are based solely on sales and recruiting additional sales consultants. There is no cap on what sales consultants earn. As sales and the recruiting of consultants rise, so do commissions.

What is interesting about Mary Kay is how many features of the firm's culture reinforce each other in a consistent manner. For example, stories of role models are prevalent throughout the organization. Almost every employee knows the story of Mary Kay Ash, who started out as a young saleswoman for Stanley Home Products. She was so poor that she had to borrow \$12 to travel from

her Houston home to Stanley's 1937 convention in Dallas. Through hard work, she built the Mary Kay Cosmetic Company and amassed a family fortune of over \$300 million. This story and those of other successful sales consultants permeate the organization, reinforcing the firm's architecture, and helping to motivate hard work and increased sales. The company is also famous for lavishly rewarding its successful sales consultants in a very public manner. The annual sales meeting is an extravaganza where individuals are crowned as "queens" and rewarded with complementary pink Cadillacs, jewelry, color-coded suits, badges, and emblems.

The message at Mary Kay is clear. Success is measured by sales and recruiting efforts. Do these things well and you will be rewarded, both financially and through public recognition.

Suggested by A. Farnham, "Mary Kay's Lessons in Leadership," *Fortune*, September 20, 1993.

A focus on organizational architecture helps identify the role of corporate culture in creating value. Indeed, our definition of architecture corresponds to some key aspects of what is often called corporate culture. For example, a company's architecture specifies how authority (or decision rights) is distributed among employees and how rewards are determined. The advantage of this economic approach is that it helps managers identify the key components of a firm's corporate culture and analyze how they might affect culture in purposeful ways.

Most companies do not write down all the features of their organization in detailed procedures manuals. Rather, much of the structure is communicated to employees in less formal and potentially less costly ways. In this sense, corporate slogans, role models, and social rituals can be viewed as efficient, cost-effective methods of *communicating* important aspects of the organization to employees. For example, a slogan like "at Ford, quality is *Job One*" emphasizes that workers are expected to focus on quality and that this focus will be rewarded by the firm. Similarly, social rituals such as training sessions and company parties can help to disseminate information by increasing the interaction among employees who might not see each other on a frequent basis. Singling out role models or heroes for special awards is another way of credibly communicating what the corporation values.

"Softer" features of organizations such as rituals and role models can be effective in reinforcing and communicating the goals of the firm; and they thus have the potential to provide important aspects of a coherent organizational architecture. Their effectiveness in certain cases has led some management experts to claim that a productive corporate culture can be molded without paying any attention to formal evaluation and compensation schemes. (Indeed, some argue that incentive-based pay is actually detrimental to a productive organization.) But it is a mistake to think of the "hard" and "soft" aspects of the organization as mutually exclusive or in competition with each other; both can play a valuable, if not an essential, role in increasing firm value.

Conclusion: Survival of the Fittest

The fundamental economic problem facing individual firms (as well as entire economic systems) is trying to ensure that decision-makers have the relevant information to make good decisions and that these decision-makers have incentives to use information productively. In addressing this problem, all firms develop what we call an organizational architecture. There are three important components of organizational architecture:

- the assignment of decision rights (rights to decide and take actions) among individuals,
- the performance evaluation system, and
- the reward system.

The factors that are likely to be particularly important determinants of a firm's architecture are *market conditions, technology, and the regulatory environment*. In heavily regulated firms—or in smaller firms in fairly stable industries—top management is likely to have more of the relevant information for decision-making. In such cases, centralized decision-making is more likely to be adopted with relatively less emphasis on formal performance evaluation and pay-for-performance systems. By contrast, in larger companies—particularly those in unregulated industries or facing stiff foreign competition—decision rights are more likely to be decentralized among managers and employees. In such cases, performance evaluation and reward systems must be designed to control agency problems and promote better decision-making.

Like the process of “natural selection” in the theory of evolution, economic competition determines which organizations survive by weeding out inefficient firms. This principle of Economic Darwinism implies that companies increase their chance of survival by adapting their organizational architecture to changes in their particular economic environments. The concept of survival of the fittest has important implications. First, existing architectures are not random; there are sound economic explanations for the dominant organization of firms in most industries. Second, surviving architectures at any

point in time are optimal in a relative rather than an absolute sense; that is, they are best among the competition, though not necessarily the best possible. These two observations together suggest that, although improvements in architecture are certainly always possible, a manager should not be too quick to condemn the prevailing organization without careful analysis. Before undertaking major organizational changes, top managers should thus have a good understanding of how the firm arrived at its existing architecture and, more generally, of why particular types of organizations work well in particular settings. Such an understanding is important if only because the costs of organizational change can be so large.

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